Predicting AI-System Correctness on Unseen Benchmarks: Calibrated Hierarchical Adaptation and the Limits of Label-Only Prediction

Satyajeeth Suresh Kannan
Team BHopper
Innovation Academy, Alpharetta, Georgia, USA
jeetusk08@gmail.com

Abstract

We predict whether an AI system answers an item of a never-seen benchmark correctly, before and after a few of its outcomes are revealed. Our predictor is a hierarchical Bayesian logistic model with a benchmark level, a system effect, an item effect and item-feature effects, fitted by MAP on every revealed label of the benchmark, with an item-text prior and a system prior from public data. It needs only `numpy` and makes thousands of predictions in about a second. Three findings matter more than the model class. First, our own formative feedback shows that the hidden benchmarks differ from the public ones: pair accuracies are spread almost uniformly and benchmark levels vary far more than in the public data, which sets one prior width. Second, a faithful re-implementation of the streaming evaluation and a simulator calibrated to the hidden pairs let us choose models without spending leaderboard uploads. Third, we locate the limit of label-only prediction: for a system alone on its benchmark our predictor is within 0.0003 ALC of the Bayes-optimal level-only estimator, and the remaining 0.024 ALC gap to an oracle that knows each pair's accuracy needs item-level knowledge that item text, sentence embeddings, small language-model states and perplexity did not supply across benchmarks. We also show that practice-board differences largely track how many times a team uploaded, because each upload is scored on 9 sampled pairs and the board keeps a team's best score.

1 Introduction

The competition [5] asks for a program that predicts $P(\text{correct})$ for an AI system s on an item i of a benchmark that was not in the training pool, first with no outcomes of that system on that benchmark and then after 1, 3, 7, 15 and 31 of them have been revealed. Predictions are scored by Brier score [1] at each budget. The setting is close to item response theory [2] and to efficient evaluation with few items [3], with two differences that drive everything below: the benchmark is new, so item parameters cannot be learned in advance, and the evaluation is streaming, so a method has to adapt from very few labels.

Our contributions are:

- A numpy-only hierarchical predictor (Section 4) that is calibrated for the hidden test's structure, which we infer from our own formative feedback without identifying any hidden benchmark (Section 3).
- Three model-selection tools that replace leaderboard probing (Section 5), including a faithful re-implementation of the official streaming protocol.

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- A measured limit: for a lone system the predictor is at the Bayes limit of label-only prediction, and the remaining gap needs item-level knowledge that did not transfer across public benchmarks in any form we tested (Sections 6 and 8).
- An analysis of what the practice board measures (Section 7).

Selected submission. Version v6 (`paie_v6.zip`, SHA-256 `45c6596a...7aea08`, full hash in Appendix A), uploaded to Codabench on 28 September 2026. Its Codabench submission ID is given in the OpenReview form. Version v7 (Section 6) adds an acquisition rule with a consistent but negligible gain; it is not selected because it has not been run on the platform, and a gain of at most 0.0005 ALC does not justify that risk.

2 Task, interface and metric

The program is called as `predict([subject, item], labeled)`. The `subject` has eight string fields (`normalized_name`, `provider`, `release_date`, `access_date`, `harness`, `harness_version`, `reasoning_effort`, `subject_features_extra`). The `item` has `item_content`, `item_features`, `interactors` and an anonymous benchmark ID; there is no reference answer and no benchmark name. `labeled` holds every outcome revealed so far, for all sampled subject–benchmark pairs.

Each pair with at least 80 items is split 50/50 into an acquisition pool and an evaluation pool, independently for each pair. An optional acquisition function sees candidates in a fixed random order; without one, labels are drawn at random. Predictions on the evaluation pool are scored by Brier score B_k at budgets $k \in \{0, 1, 3, 7, 15, 31\}$ and combined as

$$\mathrm{ALC} = 0.1\, B_0 + 0.2\, (B_1 + B_3 + B_7 + B_{15}) + 0.1\, B_{31}, \tag{1}$$

averaged equally over pairs (lower is better). Evaluation workers are recreated at each budget, so no state survives between budgets; this costs nothing because `labeled` is complete in every call. When two systems share a benchmark, one system's labels can fall on the other's evaluation items. This is the only source of item-specific evidence inside the evaluation.

3 Data, and what the hidden test looks like

Public training pool. `measurement-db` [4] has 6 benchmarks and 227,995 system–item pairs with up to 200 trials per pair: matharena (104 systems, 1,755 items), mmdocrag (68, 2,000), multi_swebench (82, 2,078), real_webagents (33, 233), researchcodebench (31, 212) and swe_rebench (1, 6,271).

Where the predictable variance is. Table 1 compares a constant equal to the benchmark mean with three oracles: the true per-system accuracy, the true per-item accuracy, and an additive logistic (Rasch) fit. On five of six benchmarks, knowing item difficulty removes more error than knowing system strength.

Table 1: Oracle Brier scores per public benchmark (trial-level, in-sample on the full data; Rasch: additive logistic fit, ridge 0.01; lower is better).

Benchmark	Benchmark mean	Subject oracle	Item oracle	Rasch oracle
matharena	0.210	0.180	0.109	0.075
mmdocrag	0.173	0.132	0.148	0.101
multi_swebench	0.128	0.106	0.068	0.057
real_webagents	0.234	0.186	0.134	0.094
researchcodebench	0.228	0.205	0.130	0.090
swe_rebench	0.249	0.249	0.045	0.046

The hidden test, inferred from our own formative feedback. Every model predicts 0.5 at budget 0, so the one-bin budget-0 calibration error equals $|0.5 - \bar{y}|$, which gives each scored pair's accuracy up to reflection. Our three scored uploads cover 26 pairs on 7 hidden benchmarks. Each sample has

9 pairs: every benchmark appears once and two of them have a second system; no system appears on two benchmarks within a sample. We find that (i) pair accuracies are spread like Uniform(0,1) ($\mathbb{E}[\bar{y}(1-\bar{y})] = 0.162$, against 0.167 for a uniform distribution); (ii) hidden benchmarks differ from each other much more than public ones (rms logit accuracy 1.70 vs. 1.08); and (iii) systems within a hidden benchmark are similar (sd 0.2–1.1 logit including sampling noise, 0.2–0.7 on 6 of the 7 benchmarks). These numbers are used only to set one global prior width and to build a synthetic validation set. No model uses a benchmark-specific value, and we made no attempt to identify the benchmarks behind the anonymous IDs.

4 Method (submission v6)

The model is

$$\operatorname{logit} p(s,i) = \beta_b + \delta_s + \varepsilon_i + G_i^\top g, \tag{2}$$

fitted per benchmark b by three block-Newton MAP sweeps on all labels revealed for that benchmark. Every system that shares the benchmark pools its labels, and items are matched by their full visible description. The priors are listed in Table 2.

Table 2: Priors of the v6 model.

Term	Prior	Detail
Benchmark level β_b	$\mathcal{N}(0, \tau_b^2)$, $\tau_b = 1.4$	wide, because hidden benchmark levels are spread
System effect δ_s	$\mathcal{N}(m_s, \tau_s^2)$, $\tau_s = 0.8$	m_s: half the system's relative strength on public data[a]
Item effect ε_i	$\mathcal{N}(m_i, \tau_i^2)$, $\tau_i = 1.5$	m_i: hashed word uni+bigram TF-IDF dual ridge[b]
Item features g	$\mathcal{N}(0, \tau_g^2)$, $\tau_g = 0.5$	one-hot `item_features` tokens, z-scored numerics[c]

[a] Logit accuracy minus the benchmark mean, matched by normalised model name; fallback: a ridge on metadata (release date, provider, size and variant tokens, reasoning effort). Name-matched systems recur across public benchmarks with strength correlations of $+0.23$ to $+0.90$. [b] Fitted to the labelled items' effects of the same benchmark and refitted three times. [c] Tokens shared by at least two labelled items, plus a slope per numeric key clipped to ± 3.

At budget 0, or when the benchmark has no label, the prediction is 0.5 (Section 8 explains why nothing better transferred). Soft labels are used as given; malformed input never raises, and the fallback is a smoothed pair mean, then 0.5. The code uses `numpy` and the standard library only and is deterministic: 3,000 predictions with 1,860 labels on ∼100 KB items take 1.1 s on a laptop CPU.

5 Choosing models without the leaderboard

1. **Leave-one-benchmark-out harness** on the public data (fractional labels, 3 seeds), with four regimes: *single* (each system alone), *mix2*, *mix8* and *pooled* (all systems of the held-out benchmark).
2. **Faithful streaming simulator.** It drives any submission through the official call shapes, reconstructed from the organisers' ingestion code: per-pair splits, the evaluator's own random acquisition rule, worker re-creation at each budget, binary labels, and a per-fold subject prior rebuilt without the held-out benchmark. Regimes: *practice* (1–2 systems of the held-out benchmark plus one filler pair from every other benchmark) and *kN* (N systems of the held-out benchmark).
3. **Hidden-calibrated simulator.** It rebuilds our three dev samples, or fresh samples with the same structure, from each pair's accuracy. Item effects follow a Rasch model (τ_{true} 1.0–2.0) shared by the systems of a benchmark. Item texts are random, so there is no text signal.

6 Results

Development history. Table 3 shows the leave-one-benchmark-out harness. The single regime is about 0.025 ALC harder, because item difficulty can only come from the system's own labels. (This harness omitted a $-\beta/\tau_b^2$ prior term that the shipped v5 contains; re-measured as shipped, v5 scores 0.1874 single and 0.1627 pooled.)

From v5 to v6. v6 raises τ_b from 0.8 to 1.4 (so a lone system's benchmark level adapts faster), adds the item-feature group effects and hardens the code. Table 4 shows that it is clearly better for a

Table 3: Leave-one-benchmark-out ALC on the public data (lower is better).

Model	Pooled	Single
Organisers' empirical-mean baseline (replicated)	0.216	0.216
Pooled hierarchical, flat prior (v1)	0.169	–
+ item-text prior (v2)	0.161	0.186
+ name-matched system prior, MAP fit, $\tau = 1.0$ (v3)	0.161	0.188
$\tau = 0.8$, system prior $\times 0.5$ (v5)	0.162	0.186

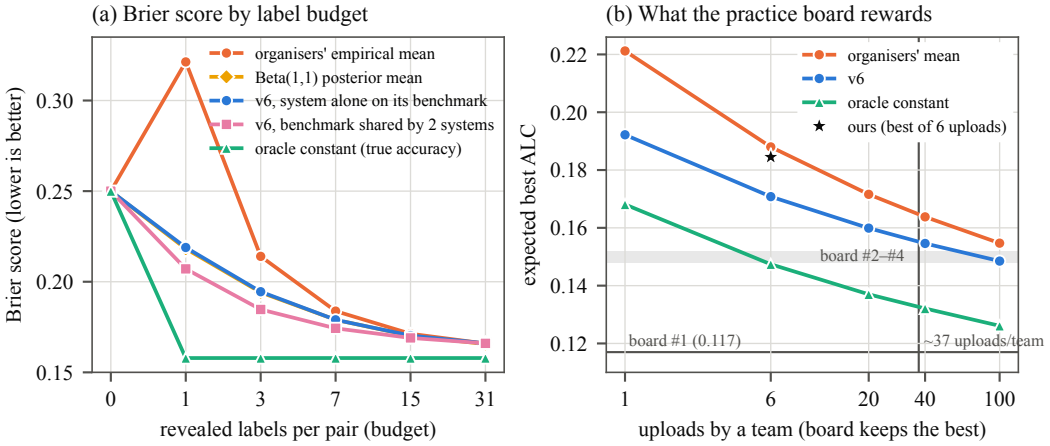


Figure 1: Hidden-calibrated simulator, 1,800 fresh 9-pair samples. (a) Brier score at each label budget; the Beta(1,1) curve lies under the v6 lone-system curve. The unsmoothed empirical mean loses most at B_1, where one label sets its prediction to 0 or 1. (b) Expected best practice-board score of a team that uploads N times, with the board as read on 7 October 2026 (#1, the band of #2–#4, our best of 6 uploads, and the mean number of uploads per team).

lone system, which is the practice-board case, and about equal when many systems pool their labels. Not every benchmark improves: researchcodebench is slightly worse in every harness regime (single $+0.0016$), and in the pooled regime v6 improves only 2 of 6 benchmarks.

Table 4: v5 (previous submission, as shipped) vs. v6 in the three tools (ALC, lower is better).

Tool and regime	v5	v6
Harness: single (3 seeds) / mix2	0.1874 / 0.1841	**0.1854 / 0.1828**
Harness: mix8 / pooled	0.1770 / 0.1627	0.1772 / 0.1634
Faithful simulator: practice (pair-weighted)	0.2051	**0.2024**
Faithful simulator: k1 / k2	0.2128 / 0.2068	**0.2111 / 0.2062**
Faithful simulator: k4 / k8	0.1991 / 0.1916	**0.1987 / 0.1915**
Hidden-calibrated: practice samples (τ_{true} 1.5)	0.1957	**0.1918**
Hidden-calibrated: 8 systems per benchmark (sd 0.4 / 0.8)	0.1778 / 0.1782	**0.1752 / 0.1765**

For a lone system, v6 is at the Bayes limit of label-only prediction. After k labels, a lone system's labels say only "m of k correct". Without item-level knowledge every unlabelled item looks the same, so the Bayes-optimal prediction is the posterior mean of the pair accuracy; with the near-uniform hidden accuracy distribution this is close to the Beta(1,1) posterior mean $(m+1)/(k+2)$. Figure 1a and Table 5 show that v6 matches it on lone pairs (a more U-shaped Beta(0.7,0.7) prior is not better), and that v6's whole gain over Beta(1,1) comes from pairs that share a benchmark, where the second system's labels inform the level and, through shared item effects, the very items being predicted. Per sample, v6 beats Beta(1,1) by 0.0027 ± 0.0028, and does so on 85% of samples.

Coordinated acquisition for shared benchmarks (v7, not selected). Splits are independent per pair, so an item that system B has revealed is in B's acquisition pool and can never inform B's scored items; if system A reveals the same item, A's label adds nothing for B. An unrevealed item,

Table 5: v6 vs. the Beta(1,1) posterior mean (ALC; hidden-calibrated simulator).

Pairs	v6	Beta(1,1)
Lone pairs, three dev samples	0.2004	0.2001
Lone pairs, fresh samples	0.1941	0.1939
Shared pairs, three dev samples	**0.1788**	0.1860
Shared pairs, fresh samples	**0.1886**	0.1949

by contrast, sits in B's evaluation pool half of the time. v7 adds a stateless `labeling.py` that skips a candidate another system of the same benchmark has already revealed, otherwise reproduces the evaluator's random rule exactly, and takes every candidate when the 31-label allowance needs them. It never looks at outcomes, so each pair's labels remain a random sample of its own acquisition pool. The gain is consistent but small: faithful simulator -0.00021 ± 0.00011 (practice), $+0.00010$ (2 systems per benchmark), -0.00029 (4) and -0.00045 ± 0.00019 (8); hidden-calibrated -0.0001 to -0.0005. A random label is already near-optimal for estimating a pair's own level; coordination can only redirect which items the other systems learn about.

7 What does the practice board measure?

The limit. The oracle constant knows each pair's accuracy from the first label and scores 0.168 on hidden-like samples, 0.024 below v6. B_0 is fixed at 0.25 for every method without prior knowledge of the pair. Any further gain requires knowing which items a system will solve.

The board. The practice board scores each upload on one fresh 9-pair sample and keeps a team's best upload. In our simulator one upload has sd ≈ 0.016 for every method. On 7 October 2026 the public Codabench API listed 1,185 submissions from 138 participants, about 37 per team on the 32-team board. Figure 1b and Table 6 give the expected best-of-N scores. Board scores of 0.148–0.152 are what a good label-only method, or even the baseline, reaches with 40–100 uploads; our 0.1845 is the best of 6 uploads of different versions. A score of 0.117 is below the oracle constant's best of 100, which implies genuine item-level knowledge or an extreme outlier. The final ranking uses one common test subset, so comparisons there are paired and much less noisy: what counts is each method's expected score, not its best upload.

Table 6: Expected best-of-N practice-board ALC in the hidden-calibrated simulator.

Method	1 upload	best of 6	best of 20	best of 40	best of 100
Organisers' empirical mean	0.221	0.188	0.172	0.164	0.155
v6	0.192	0.171	0.160	0.155	0.149
Oracle constant	0.168	0.147	0.137	0.132	0.126

8 What did not work

All results are leave-one-benchmark-out on public data unless stated otherwise.

Cross-benchmark item-difficulty priors. Using the TF-IDF ridge as a cross-benchmark prior hurts (pooled ALC 0.161 $\rightarrow$ 0.178). Table 7 tests whether richer item representations predict within-benchmark difficulty on a held-out benchmark (Spearman correlation, ridge, equal benchmark weights). No representation has a consistent sign across benchmarks; LM perplexity tracks difficulty on competition mathematics but not on software or web tasks.

Other negative results.

- Zero-shot LLM difficulty rating (Qwen3-VL-8B-Instruct, 4-bit, asked for the probability that a typical current system solves the item; 1,645 items): within-benchmark Spearman -0.16 to $+0.20$.

Table 7: Held-out within-benchmark Spearman correlation between predicted and true item difficulty.

Representation	matharena	mmdocrag	multi_swe	webagents	rcbench	mean
Text traits (length, code/math markers)	$+0.24$	-0.03	-0.04	$+0.12$	-0.32	-0.01
bge-small sentence embedding	-0.26	$+0.03$	-0.02	-0.06	-0.21	-0.11
bge-base sentence embedding	$+0.11$	-0.08	-0.07	-0.03	$+0.16$	$+0.02$
Qwen2.5-Coder-1.5B, mean last layer	$+0.07$	-0.00	$+0.08$	-0.02	-0.12	$+0.00$
Qwen2.5-Coder-1.5B, last token	$+0.17$	$+0.01$	$+0.16$	$+0.01$	-0.31	$+0.01$
Qwen2.5-Coder-1.5B, perplexity + length	$+0.37$	$+0.03$	-0.09	$+0.15$	-0.02	$+0.09$

- Sentence embeddings instead of TF-IDF within a benchmark: no gain (0.1608 vs. 0.1608 pooled).
- Exact public-item lookup: no hidden item matched a public item; budget-0 predictions stayed at 0.5 on the board.
- Non-random acquisition in the harness (most-shared, rarest, farthest-point or longest items first): all worse than random (0.171–0.180 vs. 0.165). A stateful version of the v7 rule: no gain over the stateless one (k8: 0.1738 for both).
- Prior shape and width (τ_b 1.2–2.0, τ_s 0.5–1.0, τ_i 1.2–1.8, probit moderation of the MAP prediction, a Beta(0.7,0.7) level prior): all within ± 0.0005 of v6 on hidden-like data. A system-count rule for τ_s helps on heterogeneous public systems (pooled 0.1634 $\rightarrow$ 0.1612) but hurts on homogeneous hidden-like systems (0.1634 $\rightarrow$ 0.1650), so it is off. All post-hoc calibrators failed nested validation.

9 Discussion and limitations

Without labels from other systems on the same items, we could not predict per-item difficulty on a new benchmark from item text in this data. The gains then come from calibrated shrinkage of the system's own labels, and v6 is at the Bayes limit for that task. Pooling across systems that share a benchmark is the one label-based source of item-level evidence; it is worth 0.006–0.007 ALC on such pairs, and coordinating acquisition to increase it adds at most 0.0005.

Limitations. Validation rests on 6 public benchmarks and on a synthetic model of the hidden structure built from 26 pairs. The hidden simulator is in-sample for the accuracy distribution (one global width was tuned with it, though the optimum is flat for τ_b 1.4–1.8), and its items carry no text signal. The structure of the final common test subset (number of pairs, systems per benchmark) is not published; if many systems share each benchmark, v6's advantage over v5 shrinks to zero and pooling dominates. We could not test large open models: the rules allow bundled models of up to 300B parameters, but our hardware cannot run them and the evaluation GPU and preinstalled packages are not published. Budget 0 (10% of the metric) stays uninformative.

10 Reproducibility, code and disclosure

Code. The submission, the three model-selection tools and every experiment script are released at `https://github.com/JeetuSK0808/paiec-bhopper` (MIT licence; no data is redistributed). Appendix A lists the commands behind the tables and the figure. The submission needs Python 3.12 and `numpy`; the experiments also use `pandas` and `pyarrow`, the LM probes `torch` and `transformers`, and the figure `matplotlib`. Everything ran on one Apple-silicon laptop. The hidden-calibrated simulator also reads one small table derived from our own formative-feedback logs (per-pair accuracies of the hidden pairs we were scored on). It is not redistributed, because the rules forbid sharing private test data; it is available to the organisers on request.

Disclosure. Training data: `measurement-db` only (the public training pool); no other benchmark data, and no per-item results from outside the pool. Pretrained models: none in the submission. BAAI/bge-small-en-v1.5, BAAI/bge-base-en-v1.5, Qwen/Qwen2.5-Coder-1.5B and Qwen3-VL-8B-Instruct (4-bit) were used only in the negative-result experiments of Section 8. Formative feedback: only our own three scoring logs, used only for the global calibration in Section 3. AI

assistance: an LLM-based coding assistant (Claude Code) was used during development for code, experiments and drafting; all numbers were produced by the released scripts.

References

[1] G. W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.

[2] G. Rasch. *Probabilistic Models for Some Intelligence and Attainment Tests*. Danish Institute for Educational Research, Copenhagen, 1960.

[3] F. M. Polo, L. Weber, L. Choshen, Y. Sun, G. Xu and M. Yurochkin. tinyBenchmarks: evaluating LLMs with fewer examples. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

[4] AIMS Lab, Stanford University. `aims-foundations/measurement-db` (gated dataset), Hugging Face, 2026. https://huggingface.co/datasets/aims-foundations/measurement-db

[5] AIMS Lab, Stanford University. The Predictive AI Evaluation Competition, 2026. https://aims.stanford.edu/competition

A Commands behind the results

Selected submission v6: `submissions/paie_v6.zip`, SHA-256 `45c6596a2631b4ed714916f0da35a2461b8e53c910867f816ee6b55bda7aea08`.
Variant v7 (not selected): `submissions/paie_v7.zip`, SHA-256 `14a30de8ce82527bf1a48d6ceec2c74297e45b5c7b7eaf1e9a50e72470ed7436`.

Table 3 comes from earlier versions of the leave-one-benchmark-out harness (`scripts/item_prior_cv.py` and its predecessors); the shipped v5 is re-measured with the current tools in Table 4. All commands below run from the repository root with the prefix `nice -n 10 env VECLIB_MAXIMUM_THREADS=1 OMP_NUM_THREADS=1 python -B`.

```
# oracle Brier scores (Table 1)
scripts/oracle_table.py
# v6 in the three tools (Table 4)
scripts/exp_synth_v6cv.py --sub submission_v6 --regimes single,mix2,mix8,pooled \
    --seeds 0,1,2
scripts/exp_synth_sim.py --sub submission_v6 --regimes practice --episodes 20 --prior fold --fast
scripts/exp_synth_sim.py --sub submission_v6 --regimes k1,k2,k4,k8 --episodes 12 --prior fold --fast
scripts/exp_synth_hiddensim.py --episodes 60 --baselines \
    --cfgs '[["v6", {"sub": "submission_v6"}]]'
# lone vs shared pairs, Bayes limit, best-of-N (Tables 5-6, Figure 1)
scripts/exp_v7_hsplit.py --episodes 60 --seeds 0,1,2
scripts/exp_v7_hsplit.py --boot --episodes 200 --seeds 0,1,2
python report/make_figures.py report/fig_curves.pdf
# coordinated acquisition (v7)
scripts/exp_v7_acq.py --regime practice --episodes 60 --seeds 0,1,2 \
    --policies random,skipco,skipseen
scripts/exp_v7_sim.py --sub submission_v7 --regimes practice --episodes 20 \
    --prior fold --fast --seed 0
scripts/exp_v7_robust_labeling.py submission_v7
# transfer probes (Table 7)
scripts/exp_v7_xfer_emb.py
scripts/embed_items_lm.py Qwen/Qwen2.5-Coder-1.5B
# packaging check
python upstream/paiec_baseline/check_submission_zip.py submissions/paie_v6.zip
```